

Second Graders as Spellers: What Types of Errors Are They Making?

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Abstract

Beginning second-grade students ($N = 60$) were administered a researcher-developed, dictated spelling test. Spelling errors of students were analyzed by grade-level spelling patterns and linguistic characteristics: phonological, orthographic, orthographic image, transposition, and morphological. Results revealed that morphological spelling pattern errors occurred most frequently. The frequency of linguistic category errors for this sample, ranging from least to most, was transposition, phonological, orthographic image, orthographic, and morphological. Further analysis of the bottom quarter of students, as determined by their performance on a timed word reading measure, indicated a pattern of spelling errors similar to the rest of the sample, but occurring in significantly greater quantity on average. These findings support previous research suggesting poor spellers display similar error patterns as better spelling peers, but with higher frequency. Initial implications suggest to educators that in spite of the wide range of spelling ability in their classrooms, the pattern of knowledge appears similar. Students rely on multiple sources of linguistic knowledge, and by analyzing their spelling errors based on the information students use to generate their spellings, instruction may be provided to explicitly and systematically target students' weaknesses.

Keywords

spelling, second grade students, phonological, orthographic, morphological

Spelling is a linguistic skill; it is the visual representation of spoken language and relies on one's knowledge of the phonological, morphological, and orthographic structure of the English language (Perfetti, 1997). Much of the previous research on spelling has focused on students' spelling ability on specific orthographic and morphological patterns (Apel, Masterson, & Niessen, 2004; Nunes, Bryant, & Bindman, 1997; Reece & Treiman, 2001; Treiman, Cassar, & Zukowski, 1994) and has not investigated all three linguistic components within one group of students to observe the interaction of their use of phonology, orthography, and morphology. The purpose of this study was to investigate the spelling patterns and linguistic knowledge of beginning of the year second graders and to explore the differences between good and poor spellers. Knowledge of these patterns and differences would provide educators with instructional guidance to meet the spelling and reading needs of their students.

Phonological knowledge is one's awareness of the sound system of language. It has been identified as an essential skill in both reading and spelling development (National Reading Panel, 2000; Treiman & Bourassa, 2000). Morphological knowledge is one's awareness of the units of meaning in language (Goswami, 1998). English spellings have developed over time to keep the morphological meaning intact over phonological representation. For

example, "sign" and "signal" are morphologically related, but by adding the "al," the phonological characteristics of the root word are altered. Orthographic knowledge is one's knowledge of the sound/symbol correspondences of English and the rules and generalizations of how sounds are represented in print (Apel et al., 2004; Cassar & Treiman, 2004; Treiman & Cassar, 1997). In some instances knowledge of the rules and generalizations is not enough. Take for example the word *coat*. It could also be spelled *cote*. Both are plausible orthographic spellings and are phonologically accurate, but only one is correct. Ehri (1980) proposed that spellings and pronunciations join to become an "orthographic image." This knowledge develops into a bank of words that children acquire as visual images due to repeated exposure to printed text. So it becomes necessary for children to rely on their orthographic images to accurately spell the word *coat*.

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Spelling and Reading Relationship

If spelling is a linguistic skill that requires knowledge of phonology, orthography, and morphology and reading requires knowledge of these same components, then what is the connection between reading and spelling? Ehri (2000) has found the range of correlations between spelling and reading performance to be .68 to .86, indicating a relatively strong relation. Learning to spell helps students with reading, but learning spellings from reading is not assured (Treiman, 1998). If one struggles with reading it is fairly common to also struggle with spelling, but a good speller is rarely a poor reader. When reading words, students need to be able to decode most of the letter-sound correspondences within the word to reach the correct target word. In spelling, the entire string of letters must be represented in the correct sequence to obtain the correct target word. The relationship that exists for spelling and reading is asymmetrical, with good readers able to read many more words than they can spell (Berninger, Abbott, Abbott, Graham, & Richards, 2002; Bosman & Van Orden, 1997).

Spelling gives evidence of children's knowledge of reading via phonemic awareness, orthographic knowledge, and morphological knowledge because it requires precision (Ehri, 2000; Treiman, 1998). Researchers studying the relationship between spelling and reading have found that spelling ability predicts the later reading success of students from early grades through late elementary school (Berninger et al., 2002; Mann, 1993; Treiman, 1998). Teaching students the rules and patterns of spelling will help increase both spelling and reading knowledge because advanced knowledge of the orthographic and morphological structure of English can assist vocabulary knowledge and reading comprehension. Spelling aids these higher level components of reading because words are systematically spelled to preserve morphological features before phonological features (e.g., *human*, *humanity*; Apel et al., 2004; Moats, 1995; Venezky, 1999). Because of the reciprocal relationship between reading and spelling, Conrad (2008) recommends aligning reading and spelling curricula.

Previous Spelling Research

Much of the research on the development of student spelling knowledge has focused on children in early elementary, kindergarten or first grade (Sharp, Sinatra, & Reynolds, 2008; Treiman, 1993). A few studies have explored the spelling knowledge of second graders within a range of elementary school years (Beers, Beers, & Grant, 1977; Beers & Beers, 1992; Foorman, Jenkins, & Francis, 1993; Schlagal, 1992; Walker & Hauerwas, 2006). Other researchers have investigated students' ability to spell a particular spelling pattern such as vowels versus

consonants, syllabic *r*, morphological endings (e.g., plural *-s*, past tense *-ed*), or alveolar flaps (e.g., the /t/ in *party* or the /d/ in *lady*) across a variety of grade levels (Jorm, 1977; Kemp & Bryant, 2003; Nunes et al., 1997; Reece & Treiman, 2001; Stage & Wagner, 1992; Treiman et al., 1994). In general, these findings indicate that vowels are more difficult to learn to spell than consonants, past tense *-ed* is acquired over time, and plural *-s* is relatively easy for children to represent. These are some of the patterns of development that have been explored, but further research is warranted to include the interaction of additional patterns and linguistic characteristics, in general, and for second graders specifically.

Students' Spelling Abilities

Not all students possess the same level of spelling ability. In fact, one's level of spelling ability is oftentimes used as a "red flag" for potential disabilities. Previous research has largely focused on students with dyslexia and compared them to spelling matched peers to inform what is currently known about poor spellers (Cassar & Treiman, 2004). In this literature there has been a discussion regarding whether poor spellers lag behind average spellers or exhibit a different pattern of growth (Lennox & Siegal, 1996). Thus far, the research findings have been mixed regarding the spelling performance of these groups. Some research has documented that students identified as poor spellers generate the same types of errors as matched younger children with typical spelling ability (Kamhi & Hinton, 2000; Nelson, 1980; Treiman 1997), while others have found differences (Bailet, 1990; Lennox & Siegal, 1996). Additional research is needed to explore the qualitative differences between good and poor spellers with and without a disability.

Purpose

The purpose of this study was to provide educators with information currently lacking on student spelling knowledge for patterns and linguistic characteristics as well as to explore potential spelling differences between good and poor spellers. Educators would benefit from an initial understanding of what second graders know about spelling as they start the school year because it suggests which spelling patterns should be the focus for instruction during the school year. This knowledge would also be beneficial when thinking about how to differentiate instruction for the various literacy abilities found in a classroom. Because of the relationship between spelling and reading (Berninger et al., 2002; Conrad, 2008; Ehri, 1997, 2000), educators can benefit from the ease of using a group-administered spelling test to inform instructional decisions for both reading and

spelling. Analyzing the spellings of individuals is a valuable tool because it can provide necessary information regarding their understanding and mastery of orthography, phonology, and morphology all in one assessment (Moats, 1995; Treiman, 1998), and this knowledge can be applied to reading as well as spelling instruction. For this study, it was hypothesized that morphologically based spelling patterns and orthographic image and morphological linguistic errors would be the most prevalent types of errors for all students based on the degree of knowledge they require. It was unknown whether poor spellers would exhibit different patterns of spelling errors compared to good spellers. Based on need for additional spelling information, the following three research questions were addressed:

Research Question 1: What are the most frequent errors made on a dictated spelling test consisting of standard patterns typically taught and assessed in second grade?

Research Question 2: What are the most frequent linguistic categories of the errors made on this same dictated spelling test?

Research Question 3: Are there differences in the patterns of error types for the bottom quarter of students when compared to the rest of the sample?

Method

Participants

The participants in the current study were second-grade students who had taken a researcher-designed spelling test as part of a larger research study conducted in a southeastern state with the Florida Assessments for Instruction in Reading (FAIR; see Florida Department of Education, 2009, for an explanation of the larger study, which involved a diverse sample of 647 students). The FAIR was administered in October and only data from second graders' spelling and reading subtests were of interest in the present study. To obtain the current participants, second-grade students in the larger research study were sorted by those with and without complete demographic information. Of the remaining 600 students, 10% ($n = 60$) were randomly selected for inclusion in this study. The mean age of this sample was 7.8 years. Of the students, 28 were female and 32 were male; 5 were characterized as having Limited English Proficiency (LEP) and 30 students qualified for free and reduced price lunch (FRPL). In addition, 32 students were characterized as White, 17 as African American, 9 as Latino, and 2 students were characterized as Multiracial. This sample followed a similar demographic pattern as the larger research sample, which had 52% male, 35.7% White,

26.9% African American, and 16.4% Latino. Forty-eight percent were characterized as FRPL and 8.7% as LEP.

Materials

Spelling. A 25-word dictated spelling test was used to assess spelling ability (see appendix for words, patterns, and linguistic characteristics). This test was developed after identifying second-grade spelling patterns contained in the curriculum standards of states that included spelling in their standards, the scope and sequence of the most frequently used core reading programs in the state, supplemental and intervention reading and spelling programs, and previous research findings on spelling development (Henry, 2003; Schlagal, 1992; Treiman, 1993). Based on the instructional expectations in the standards and published programs, words representing the spelling patterns were generated and organized into the word list used for this study. The process of spelling pattern and word selection was similar to what was used by Foorman and Ciano (2005). They explored the spellings of third-grade students using a spelling inventory with words that were selected based on spelling patterns deemed grade level by state standards.

The spelling patterns selected for assessment fell within the following categories: short vowels, long vowels containing a single letter, vowel team (two or more graphemes and includes diphthongs), Floss Rule (one syllable word with a short vowel that ends in *f*, *l*, or *s* is spelled *ff*, *ll*, *ss*), Changing Rule (if a consonant is before a final *y*, change the *y* to *i* when adding a suffix that does not begin with *i*), the Doubling Rule (if a word ends in one vowel and one consonant and the final syllable is accented with a vowel suffix being added, then the final consonant is doubled), past tense-*ed*, consonant digraphs, *s* blends in the initial position, irregular plural nouns (e.g., tooth/teeth; Henry, 2003), plural /*s*/ and /*z*/, vowel-consonant-*e*, *r*-controlled, and *ck* spelling pattern (see appendix for words representing patterns). The labels used to represent these spelling patterns may vary by reading or spelling program (e.g., the vowel-consonant-*e* pattern is also referred to as the silent *e* or the magic *e*), so for the purposes of categorization in this study the aforementioned labels, unless otherwise cited, were used and defined according to Carreker (2005) as representative of the typical labels for the spelling patterns selected for second grade. It is important to note that while these labels are instructionally relevant, there is often overlap between them and the categorical distinction is sometimes arbitrary (e.g., past tense-*ed* words could also be grouped in the Doubling Rule category).

The standard frequency index (SFI) from *The Educator's Word Frequency Guide* by Zeno, Ivens, Millard, and Duvvuri (1995) was used to determine the frequency of the words. For instance, a word with an SFI of 65.0 has a

frequency per million that is 10 times higher than the frequency per million of a word with an SFI of 55.0. *The Educator's Word Frequency Guide* is a written word frequency count based on over 17,000,000 tokens (i.e., total number of words) and 164,000 types (i.e., number of different words), spanning texts from kindergarten through college. Words selected for this test fell within a SFI range of 47.0 to 70.9.

Reading. A timed word reading measure (Florida Department of Education, 2009) was used to group students into good and poor readers. The timed word reading measure required students to read as many words as possible in 45 seconds from a list of 80 words. The poorest performing students on the reading measure were grouped into the bottom quarter and compared with the remaining students on spelling performance. This word reading measure was used to form groups because previous research findings have noted the reciprocal relationship between word reading and spelling (Ehri, 1997, 2000). In this sample, performance on the word reading measure was highly correlated with performance on the spelling test, $r = .83$, $p < .01$, and allowed for separation between the independent and dependent variables.

Design and Procedure

A descriptive group design was used to answer the research questions. The spelling test was group administered in students' classrooms in the following format: A research assistant said the word, provided a sentence containing the word, and said the word again. Students were given a pencil and a student record sheet that was numbered and lined. Individual spelling tests were then entered into a spreadsheet using researcher-assigned student identification numbers, the target word, and student spelling of the word. Frequency counts for the student spelling of each word (correct or incorrect) were calculated and responses were further analyzed by coding for spelling pattern error and a specific linguistic error type (see Data Coding System). Students were then sorted by quarter based on their performance on the timed word reading measure and divided into two groups: the bottom quarter and the remaining 75% of students.

Data coding system. The first author, a doctoral student with training as a speech-language pathologist and reading specialist, coded words students were asked to spell correct or incorrect according to the target spelling patterns for which that word was included in the spelling list (e.g., short vowel or consonant digraph; see appendix). Student errors were also organized into five categories designed to tap into the underlying linguistic components of each target word. Each phoneme of the target word was analyzed for a linguistic error type. Therefore, each word had the potential to

generate multiple errors from more than one of the five categories used (see appendix). The following categories and their operationalized definitions were modeled after the previous work of Masterson and Apel (2000) and Treiman (1993): (a) phonological errors: a phoneme is not represented by a grapheme(s) in the spelling of a word (e.g., *do/dog*); (b) orthographic errors: letter(s) is used to represent a phoneme in a word that is not possible in English orthography; a spelling rule is not applied when required (e.g., *bick/back*); (c) orthographic image errors: legal/plausible but incorrect representation of the phoneme in that particular word (e.g., *boyl/boil*); (d) morphological errors: the prefix or suffix is omitted, misspelled, or when the suffix is added, the needed modification to the base is not spelled accurately (e.g., *planing/planning*); (e) transposition errors: the correct representation of phonemes is selected but two adjacent phonemes occur in the wrong letter sequence (e.g., *nad/and*).

Interrater Reliability

Of the data, 20% were coded for interrater reliability. Point-by-point interrater reliability of 96% was obtained for identifying the target spelling patterns. Point-by-point interrater reliability of 92% was obtained for linguistic error types. The first author, a doctoral student with training as a speech-language pathologist and reading specialist, coded all the data. The second coder was a doctoral student who was also trained as a speech-language pathologist and reading specialist. The second coder was trained on the two coding systems used in this study, through discussion and practice with coding, until 100% agreement was achieved on practice words for both the target spelling pattern and the linguistic error types.

Results

Spelling Pattern Errors

To address the first research question, the frequencies of incorrect spellings were analyzed using the target spelling pattern. The mean total words spelled correctly for the 25-word spelling test was 12.45 ($SD = 5.93$). The range of total items correct was 2 to 24. Table 1 reports the total patterns correctly spelled out of the total opportunities given to spell each of the target spelling patterns for all students. Because the FAIR assessment procedures from the larger study set a criterion of 80% correct as a practical benchmark to help educators determine which skills were mastered and which would need additional instruction, it was felt that patterns with less than 80% correct would be of practical interest to educators as a starting point to begin planning for instruction (Florida Department of

Table 1. Frequency of Students' ($N = 60$) Correct Spelling of Target Spelling Pattern

Target Spelling Pattern	Total Correct	Total Opportunities	Percentage Correct
Doubling Rule	4	120	3
Changing Rule	7	60	12
Irregular plural noun	22	60	37
Past tense- <i>ed</i>	56	120	47
<i>r</i> controlled	60	120	50
Vowel team	324	540	60
Floss Rule	114	180	63
Vowel-consonant- <i>e</i>	43	60	72
<i>s</i> blend initial	184	240	77
<i>ck</i> pattern	95	120	79
Plural / <i>s</i> / and / <i>z</i> /	48	60	80
Short vowel	575	660	87
Long vowel	161	180	89
Consonant digraph	219	240	91

Education, 2009). The following 10 spelling patterns met this criterion: Doubling Rule, Changing Rule, irregular plural nouns, past tense-*ed*, *r*-controlled, vowel team, Floss Rule, vowel-consonant-*e*, *s* blends, and *ck* pattern.

Of these 10 spelling patterns, 5 were correctly represented 50% or less: Doubling Rule, Changing Rule, irregular plural noun, past tense-*ed*, and *r*-controlled. In addition to students having difficulty with these 5 spelling patterns it was interesting to note that the next two areas of difficulty were vowel team with 60% correct and the Floss Rule with 63% correct. Students correctly represented the vowel-consonant-*e* spelling for the long vowel sound 72% of the time. *S* blends and the *ck* pattern were the closest to 80% correct with percentages of 77 and 79, respectively.

Linguistic Characteristics of Errors

In answer to the second research question, descriptive analyses of student spelling errors by linguistic characteristic were conducted on all incorrectly spelled words. Frequency counts indicated that out of the total opportunities available to students to make one of the linguistic characteristic errors, morphological errors occurred most often, 56.2% of the time. Table 2 displays the occurrences of student spelling errors by linguistic category, the total opportunities to generate that type of error, and the percentage of occurrence. The total occurrences of linguistic errors from least to most were as follows: transpositions 26, phonological 70, orthographic image 143, morphological 236, and orthographic 466.

However, not all linguistic categories had the same number of opportunities to occur, and therefore, the percent of occurrences based on the total number of opportunities

Table 2. Student ($N = 60$) Spelling Errors by Linguistic Characteristic

Error Type	M	SD	Total Occurrence	Total Linguistic Opportunity	Percentage of Occurrence
Phonological	1.17	1.54	70	5,400	1.30
Orthographic	7.77	5.27	466	5,460	8.53
Orthographic image	2.38	1.75	143	1,740	8.23
Morphological	3.93	1.31	236	420	56.20
Transposition ^a	0.42	0.62	26	—	—

Linguistic characteristics were defined as follows: (a) phonological errors: a phoneme is not represented by a grapheme(s) in the spelling of a word (e.g., *do/dog*); (b) orthographic errors: letter(s) is used to represent a phoneme in a word that is not possible in English orthography; a spelling rule is not applied when required (e.g., *bick/back*); (c) orthographic image errors: legal/plausible but incorrect representation of the phoneme in that particular word (e.g., *boyl/boil*); (d) morphological errors: the prefix or suffix is omitted, misspelled, or when the suffix is added, the needed modification to the base is not spelled accurately (e.g., *planing/planning*); (e) transposition errors: the correct representation of phonemes is selected but two adjacent phonemes occur in the wrong letter sequence (e.g., *nad/and*).

a. Transposition errors were not calculated due to low number of occurrences.

was calculated to determine which error types were most frequently occurring. Phonological, orthographic, and transposition errors could occur on potentially any word or more than once within a word. Orthographic image and morphological errors were more restricted in the number of occurrences that were plausible. Total opportunities for transposition errors were not calculated as they only occurred on 26 occasions and could have occurred in a number of instances. Of the opportunities available for students to make linguistic spelling errors, phonological errors occurred 1.30%, orthographic errors occurred 8.52%, orthographic image errors occurred 8.23%, and morphological errors occurred 56.2%. Students rarely omitted a sound in their spellings when one was required. Even though orthographic errors were the most frequently occurring, when taking into consideration the large number of opportunities to generate this type of error, it is interesting to note they occurred less frequently than orthographic image and morphological errors. Few students added a morphological ending where one was not required, but rather it was more common to omit them. When morphological endings were represented in students' spellings, it was often without correctly altering the base (e.g., *planing/planning*).

Pearson *r* correlations were also conducted with the linguistic error types to determine the existence of a relationship between total words spelled correctly and linguistic errors for all students. Phonological ($r = -.537, p < .01$), orthographic ($r = -.965, p < .01$), orthographic image ($r = -.696, p < .01$), and morphological ($r = -.686, p < .01$) errors were significantly correlated with total words correct. Transposition errors were not significantly correlated

Table 3. Demographic Characteristics of Bottom Quarter and Remaining 75% of Students

Demographics	Bottom Quarter (<i>n</i> = 15)	75% of Students (<i>n</i> = 45)
Age (mean in years)	8.09	7.78
Gender		
Female	4	24
Male	11	21
Ethnicity		
White	9	23
African American	4	13
Latino	2	7
Multiracial	0	2
Limited English Proficiency	1	4
Free and Reduced Price Lunch	11	19

($r = -.095$, $p = .469$). An alpha level of .05 was used for all statistical tests.

Comparison of Bottom Quarter Student Errors to the Remaining 75% of Students

The third research question focused on whether there was a difference in the types of spelling errors made by the bottom quarter and the rest of the sample. One would intuit that the bottom quarter would have the most spelling errors by the fact that they are the poorest performing students of the peer group, but the data were analyzed to determine if the errors made by these students were qualitatively different. Table 3 presents the demographic information for both groups.

Students in the bottom quarter did not differ from the remaining students in terms of the pattern of spellings that were difficult to represent correctly in their spellings. They generated the most errors with the Doubling Rule (0% correct), Changing Rule (0% correct), past tense-*ed* (0% correct), irregular plural noun (20% correct), *r*-controlled (20% correct), Floss Rule (31% correct), vowel team (31% correct), vowel-consonant-*e* (53% correct), *ck* pattern (63% correct), and *s* blends (70% correct). Using the same cutoff of less than 80% correct, students in both groups made the most errors with the same spelling patterns. What is interesting to note is that students in the bottom quarter also made a relatively high percentage of errors on words containing the long vowel, single letter pattern (73% correct) and short vowels (76% correct). There were only two spelling patterns that students in the bottom quarter spelled correctly at least 80% of the time: plural /*s*/ and /*z*/ (80% correct) and consonant digraphs (82% correct).

When analyzing the linguistic knowledge of the bottom quarter compared to the remaining students, both groups produced the same distribution of transposition errors, phonological errors, orthographic image errors, morphological errors, and orthographic errors. However, when total errors out of total opportunities for each type of linguistic error

were considered, students in the bottom quarter made more orthographic errors than orthographic image errors while the remaining students generated an equal percentage of orthographic and orthographic image errors (6%). For example, the word *sheep* was more likely to be represented with an implausible letter choice as in *ship* by a student in the bottom quarter whereas a student who was not in the bottom quarter was more likely to represent the word as *sheap*. The percentage of errors for students in the bottom quarter from least to most was as follows: phonological 2%, orthographic image 13%, orthographic 15%, and morphological 74%. The percentage of errors for the remaining students from least to most consisted of: phonological 1%, orthographic image 6%, orthographic 6%, and morphological 50%. Table 4 presents the means and standard deviations of the bottom quarter and the remaining students.

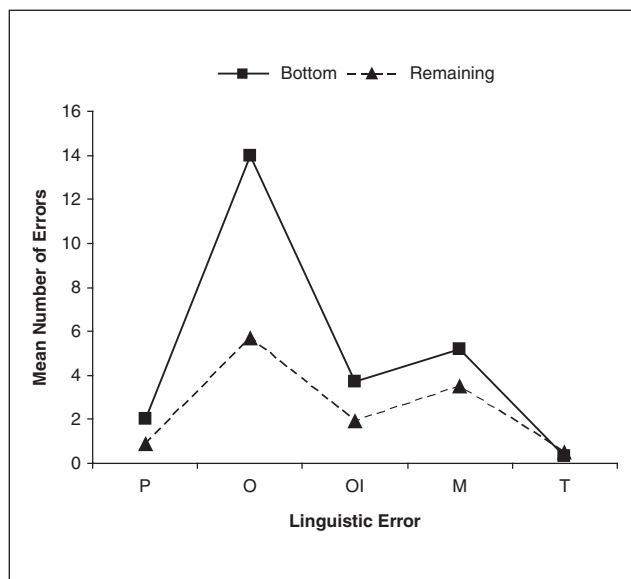
A profile analysis was performed on five dependent variables: phonological errors, orthographic errors, orthographic image errors, morphological errors, and transposition errors. The grouping variable was quartile on timed word reading (bottom quarter and remaining 75%). Profile analysis was used because it is a special application of multivariate analysis of variance when there are multiple dependent variables on the same scale (Tabachnick & Fidell, 2007). SPSS GLM was used for this analysis. Assumptions regarding normality, linearity, multicollinearity, and homogeneity of variance were met. The assumption of sphericity was violated; therefore, the Greenhouse-Geisser test was used for statistical analyses. The profiles, seen in Figure 1, deviated significantly from parallelism, $F(1.85, 107.49) = 36.28$, $p < .000$. For the levels test, statistically significant differences were found between the two groups of students when their performance was averaged across linguistic characteristic categories, $F(1, 58) = 49.00$, $p < .000$. When averaged over student groups, the linguistic characteristic categories were found to deviate significantly from flatness, $F(1.85, 107.49) = 181.98$, $p < .001$.

Student performance in these two groups was not parallel. All students in this sample generated orthographic errors the most frequently. However, the difference between the two groups of students on orthographic errors ranged from 5 to 10 times larger than any other difference except for transpositions, which had no observed difference. The difference in mean number of errors between groups was significant. Students in the bottom quarter generated more phonological, orthographic, orthographic image, and morphological errors. Figure 1 illustrates the differences in type of linguistic error. Visual inspection of the plotted profiles indicated there was no group difference on transposition errors. To further analyze the practical importance of these findings, effect sizes (ES) using Cohen's *d* were calculated for observed group differences. Algina and Keselman's (2003) method of calculating confidence intervals for ES was also used. It should be noted that these need to be interpreted with caution due to the small number of students in the bottom quarter.

Table 4. Mean Linguistic Errors of Bottom Quarter and Remaining 75% of Students

Error Type	Bottom Quarter (<i>n</i> = 15)			Remaining 75% of Students (<i>n</i> = 45)		
	Total Occurrence	<i>M</i>	<i>SD</i>	Total Occurrence	<i>M</i>	<i>SD</i>
Phonological	30	2.00	1.69	40	0.89	1.40
Orthographic	210	14.00	2.78	256	5.69	4.13
Orthographic image	56	3.73	0.96	87	1.93	1.72
Morphological	78	5.20	1.08	158	3.51	1.10
Transposition	5	0.33	0.62	21	0.47	0.63

Linguistic characteristics were defined as follows: (a) phonological errors: a phoneme is not represented by a grapheme(s) in the spelling of a word (e.g., *do/dog*); (b) orthographic errors: letter(s) is used to represent a phoneme in a word that is not possible in English orthography; a spelling rule is not applied when required (e.g., *bick/back*); (c) orthographic image errors: legal/plausible but incorrect representation of the phoneme in that particular word (e.g., *boyl/boil*); (d) morphological errors: the prefix or suffix is omitted, misspelled, or when the suffix is added, the needed modification to the base is not spelled accurately (e.g., *planing/planning*); (e) transposition errors: the correct representation of phonemes is selected but two adjacent phonemes occur in the wrong letter sequence (e.g., *nad/and*).

**Figure 1.** Profiles of linguistic category errors for the bottom quarter and remaining 75% of students.

P = phonological, O = orthographic, OI = orthographic image, M = morphological, T = transposition.

Phonological errors had a calculated $ES = .75$, 95% confidence interval (CI; 0.15, 1.35); orthographic errors $ES = 2.16$, 95% CI (1.45, 2.86); orthographic image errors $ES = 1.15$, 95% CI (0.52, 1.76); and morphological errors $ES = 1.54$, 95% CI (0.89, 2.19). All four ES are large according to Cohen (1988), indicating the performance between the bottom quarter of students and the remaining 75% was of educational as well as statistical significance.

Discussion

This study was designed to provide educators with information regarding the spelling knowledge of second-grade

students at the start of the school year. If educators are aware of spelling skills students typically bring to school, their instructional time and energy can be focused to address specific areas of need. Due to the reciprocal relationship between spelling and reading (Berninger et al., 2002; Conrad, 2008; Ehri, 1997, 2000), assessing spelling knowledge and using data gleaned from it to drive instruction is both a time-efficient and cost-effective way for educators to monitor student learning.

Spelling Pattern Errors

The first research question was asked to investigate the spelling abilities of students when compared to second-grade spelling patterns. Results indicated there were 10 spelling patterns second graders in this study frequently misspelled. The Doubling Rule, the Changing Rule, and past tense-*ed* require students to utilize their morphological knowledge. English, being a less transparent orthography (Moats, 1995), presents a particular challenge to students learning to read and spell it. The *r*-controlled spelling pattern requires students to develop accurate orthographic images for the correct representation of *er*, *ur*, and *ir* as in *skirt*. Irregular plural nouns (e.g., *geese*, *moose*) require students to understand the grammatical rules for changing from singular to plural in addition to the orthographic and morphological changes or lack thereof. Typically in English the marker *e* is added to a word ending in *s* to minimize the chance of it being confused for a plural (e.g., *please*; Moats, 2000), but in both of these examples the word is plural. In addition, the Floss Rule, vowel team, vowel-consonant-*e*, *s* blends, and *ck* spelling patterns require both orthographic and orthographic image knowledge. First, students must learn the letter(s) that represents the sounds or the spelling rule that provides the reason for representing the spelling of a particular word in that way

(e.g., doubling *f* at the end of the word *cuff*). Then, students must learn to recognize when to apply the correct version of the spelling of the same sound (e.g., when to represent the long *a* sound as *ai*, *ea*, or *igh*). According to the scope and sequences of the reviewed programs and state curriculum standards, these 10 spelling patterns are to be taught and reviewed during the second-grade year and so it is not surprising that students at the beginning of second grade struggled with them. In addition, these findings suggest it is important for educators to teach these spelling patterns during the year.

Linguistic Characteristics of Errors

The second research question proposed to examine the most frequently occurring types of linguistic errors second-grade students generate when spelling words. It was hypothesized that orthographic image and morphological errors would be the most prevalent types of errors in this sample because of the increased level of difficulty students have with morphology (Nunes et al., 1997; Treiman & Bourassa, 2000). This was not entirely supported by these findings as orthographic and morphological errors were the most frequently occurring types of errors in this sample. An orthographic error differs from an orthographic image error in that an orthographic error occurs when a letter(s) is used to represent a phoneme in a word that is not possible in English orthography or a spelling rule is not applied when required. An orthographic image error is more restricted in that a student uses a legal or plausible English representation of a phoneme, but it is incorrect in that particular word (e.g., *wayt/wait*). Considering the potential that exists to make an orthographic error on any part of each word administered, this is not surprising. When taking the total number of opportunities to make a specific linguistic error into account, morphological errors occurred the most, 56.2%, and orthographic were second with 8.53%. Orthographic image errors were very close to orthographic errors, occurring 8.23%. Treiman (1993) found with first graders that a shift from orthographic to orthographic image errors occurred between the first and second semester of school. It could be that a similar pattern would emerge in second grade. Students in this study were given fewer opportunities to make a morphological error, compared to other error types, and still this was the most frequently occurring error type. This pattern of errors would suggest that, as Treiman found with first graders, the second graders in this sample were still mastering English orthography and morphology and would benefit from explicit instruction in both.

Comparison of Bottom Quarter Student Errors to the Remaining 75% of Students

The third research question explored the differences in spelling performance of students in the bottom quarter of a timed word reading measure (which correlated highly with

spelling). When examining the demographic distribution of the two groups, the only demographic trends of interest were that the bottom quarter had a slightly older mean age, contained more students qualifying for free and reduced price lunch, and 73% was male. Research on the impact of poverty and academic achievement would have predicted more students from low socioeconomic status would be in the low performing group (e.g., Snow, Porche, Tabors, & Harris, 2007). In terms of the slightly higher mean age, it leads one to wonder if some of these students were repeating second grade. The larger percentage of males in the bottom quarter is consistent with previous research revealing a bias toward males when referring students for school identification (Shaywitz, 2003; Shaywitz, Shaywitz, Fletcher, & Escobar, 1990). Reading disability was identified in males three to four times more than females. However, when these same students were assessed for reading disability by Shaywitz and colleagues (1990), no significant gender differences were found in the prevalence of those identified with reading disability. Wagner and Ahmed (2009) found evidence for male vulnerability in reading disabilities on measures of oral reading fluency and nonsense word fluency in a large sample of students with and without diagnosed reading disabilities. Other research has suggested that the gender differences found in reading achievement may be more a function of socioeconomic status and/or early school experience (e.g., teacher interactions, expectations) than true gender differences (Entwisle, Alexander, & Olson, 2007).

Students in the bottom quarter generated the same pattern of spelling errors as the rest of the students; they just made more of them. The Doubling Rule, Changing Rule, irregular plural noun, *r*-controlled, and past tense-*ed* patterns generated the most errors for all students regardless of group status. In addition to the 10 spelling patterns all students struggled with, those in the bottom quarter for reading performance also had difficulty with the long vowel (single letter) and short vowel spellings. This observed difficulty with vowel spellings aligns with previous research findings that reported vowels were more difficult for students than consonants to spell accurately (Jorm, 1977; Stage & Wagner, 1992). This is instructionally relevant for teachers looking for guidance on what to teach their students. For the range of students assessed in this study, all could potentially benefit from explicit instruction in these 10 spelling patterns with a particular focus on vowel spelling patterns for poor readers and spellers.

Previous research findings have observed that poor spellers exhibit more of a delay in their spelling abilities than a different pattern of spelling development, but this is not conclusive (Lennox & Siegal, 1996; Treiman, 1997). In the present study, students in the bottom quarter for reading performance made significantly more phonological, orthographic, orthographic image, and morphological spelling errors when compared to the rest of the students. Beyond the quantitative differences between these two groups, it

is important to note that qualitatively, student linguistic knowledge was comparable: Both groups made the fewest number of phonological and transposition errors; morphological and orthographic knowledge was the most difficult to represent correctly in spellings. These findings are encouraging in that students demonstrated a similar degree of knowledge of spelling patterns and linguistic characteristics as their more advanced age-matched peers, based on patterns of occurrence.

At present, many educators have limited exposure to instructional materials organized to reflect students' knowledge of the phonological, morphological, and orthographic aspects of spelling. Helping educators to understand the knowledge students bring to spelling and how to determine when students are lacking in one area or another is essential to shifting the mindset beyond judging spelling as simply correct or incorrect toward viewing spelling as a diagnostic and instructional tool. Initial implications suggest to educators that in spite of the wide range of spelling ability in their classrooms, the pattern of knowledge appears similar. Students rely on multiple sources of linguistic knowledge and by analyzing their spelling errors based on the information students use to generate their spellings, targeted, data-driven instruction may be provided to match student's weaknesses and support both spelling and reading acquisition.

Conclusions

Spelling instruction in orthographic patterns and morphemes is necessary at second grade. As Moats (1995) noted, this pattern of spelling errors could be a product of the lack of spelling instruction. Hence, future research exploring the impact of instruction and spelling development is warranted. In addition to the instructional focus, one must wonder about the fidelity of implementation of the curriculum that is in use. It is interesting that there were relatively few phonological and transposition errors made in comparison to orthographic and morphological errors. Students at the beginning of second grade, even those who made many spelling errors, tended to represent all the phonemes in their spellings.

Limitations and Directions for Future Research

This study has several limitations that need to be addressed. It was descriptive in nature and therefore is only suggestive of instructional implications. Future research that employs quasi-experimental or experimental design is necessary to further explore instructional questions. A closed set of spelling words was used with students; therefore, it is limited in the range of information that could be obtained. Students only had the opportunity to demonstrate their spelling ability on the words administered. In a naturalistic

setting one could observe students' spelling ability in context or how spelling patterns are represented multiple times across sentences. Future research should explore if students acquire mastery of these spelling patterns by the end of second grade. The small number of students in the bottom quarter limits this to be exploratory in nature. Future research should replicate these findings using a larger sample, more grade levels, and the addition of a longitudinal component to determine if spelling pattern errors and knowledge of linguistic characteristics represented in students' spelling change over time.

Appendix

Spelling Words Administered

Word (Spelling Patterns Assessed)	Linguistic Characteristics Contained in Words			
	P	O	OI	M
<i>Mat</i> (short vowel)	x	x	x	
<i>Sheep</i> (consonant digraph, vowel team)	x	x	x	
<i>Fly</i> (long vowel single letter)	x	x	x	
<i>Cape</i> (vowel-consonant-e)	x	x	x	
<i>Go</i> (long vowel single letter)	x	x	x	
<i>Beginning</i> (Doubling Rule, short vowel, long vowel)	x	x		x
<i>Tries</i> (Changing Rule)	x	x		x
<i>Tall</i> (Floss Rule, short vowel)	x	x	x	
<i>Took</i> (vowel team, ck pattern)	x	x	x	
<i>Lower</i> (vowel team, r-controlled)	x	x	x	x
<i>Chew</i> (consonant digraph, vowel team)	x	x	x	
<i>Each</i> (consonant digraph, vowel team)	x	x	x	
<i>Chairs</i> (plural /s/ and /z/; vowel team, consonant digraph)	x	x	x	x
<i>Strap</i> (s blend initial, short vowel)	x	x		
<i>Stack</i> (ck pattern, short vowel, s blend initial)	x	x	x	
<i>Moose</i> (irregular plural noun)	x	x	x	
<i>Planning</i> (Doubling Rule, short vowel)	x	x		x
<i>Added</i> (past tense—ed, short vowel)	x	x	x	x
<i>City</i> (short vowel)	x	x	x	
<i>Coin</i> (vowel team)	x	x	x	
<i>Draw</i> (vowel team)	x	x	x	
<i>Stuff</i> (Floss Rule, s blend initial, short vowel)	x	x		
<i>Grass</i> (Floss Rule, short vowel)	x	x		
<i>Asked</i> (past tense—ed, short vowel)	x	x	x	x
<i>Skirt</i> (r-controlled, s blend initial)	x	x	x	

Transposition errors could occur with any word.

P = phonological, O = orthographic, OI = orthographic image, M = morphological.

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